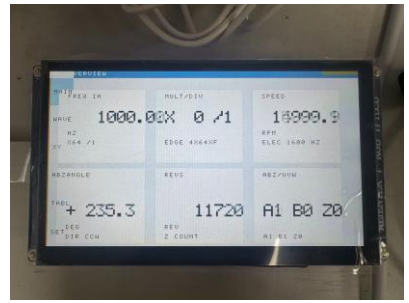
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基于FPGA的细分编码器

A 2,048-ppr magnetic encoder on a single Artix-7

16-bit interpolation · closed-form self-calibration · ABZ / UVW / SPI / UDP

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On-board 1024×600 touch UI (v62)

TEST PLATFORM

XC7A35TFGG484-2 · single 50 MHz clock

AD9238 2×12-bit, 65 MSPS front end

2,048-cycle/rev SMR ring, real turntable

1 GbE UDP capture · USB-UART 2 Mbaud

INTRODUCTION

Optical encoders degrade under dust, oil and vibration. Magnetic sensing is robust, yet the raw sin/cos bridge is never ideal: offset, gain mismatch, quadrature error and mechanical shift all fold into the angle.

Commercial interpolators fix algorithm and interface together: 10-bit resolution, opaque calibration, and multi-turn counting that needs a second die.

We implement 2,048 ppr, 16-bit angle and sub-µs latency, with ABZ, UVW, SPI and 1 GbE diagnostics from one 50 MHz clock.

TYPICAL APPLICATION CASE

Servo drives — UVW, ABZ and SPI from one 32-bit angle word.

Machine-tool rotary tables — 0.0055°/LSB at 41.5-cycle latency.

Robot joints — immunity to oil, dust and vibration.

EV traction — wide temperature range, no optical window.

KEY ENGINEERING PROBLEMS

1 Analog front end — 0.5 Vpp differential to 8 Vpp, 65 MSPS, offset binary.

2 Deterministic latency — 41.5 cycles, fixed by construction.

3 Protocol integration — ABZ 1–1024×, UVW + virtual Z, SPI, 1 GbE UDP.

4 Verification — fixed-point model, 57 benches, sign-off, real turntable.

WHY ONE ARTIX-7

Acquisition, calibration, atan2, tracking and five protocols share one clock domain. The encoder core occupies about 30% of the device; the remainder is touch UI and Ethernet diagnostics.

DESIGN FLOW

Model → RTL → 57 benches → sign-off → board

Every number traces to a CSV, log or bitstream hash.

MIT-licensed repository · 320 evidence files · 323 SHA-256 lines.

SYSTEM ARCHITECTURE

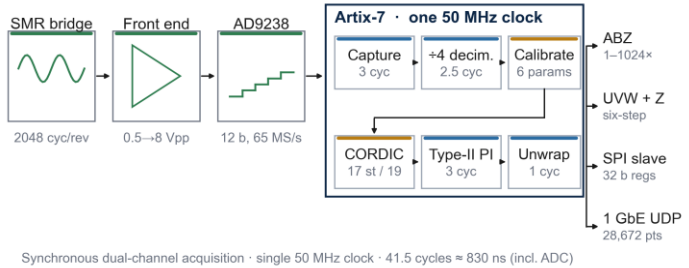


Figure 1 | Six-stage pipeline in one clock domain: 41.5 cycles ≈ 830 ns.

CREATIVE DESIGN

Closed-form calibration. A first/second-order moment solve maps the distorted ellipse to a unit circle in one pass; no LUT, no host iteration.

One clock, 41.5 cycles. No MMCM, no CDC, no soft core; latency is a design constant, not a measurement.

Wrap-safe counting and UVW. Unsigned-difference zero-extension removes the +133 → −65,403 deadlock; a wide-base branch makes the 0↔5 edge identical both ways.

HOW IT WAS PROVEN

Two silent deadlocks, found only on hardware.

Ten seconds of silence on the real turntable while 33/33 benches passed.

Build-ID bisect: probe 0x0013 reduced the failure to one step (−65,403 instead of +133); tb_acc2 reproduces 0 → 6 → 12 counts over 3,000 × +133 steps.

Five on-board criteria:

Z = 7.62–7.65 Hz, 2,048 distinct counts, 457–459 rpm, six UVW states, 60 s stream.

CREATIVE DESING

RESULT

RESULT — V90 PRODUCTION BITSTREAM

2,048	16	457–459	597,886
pulses per rev	bit angle	rpm	frames in 60 s

15.63 kHz electrical · Z 7.62–7.65 Hz · 9,964.8 fps · radius drift < 0.02%

IMPLEMENTATION

10,978 LUT (52.78%) · 8,724 FF · 32/50 BRAM · 17 DSP

WNS +0.199 ns · WHS +0.017 ns · 0 failing endpoints (24,691)

DRC 0 errors · single clock, no MMCM · 1,504,220 B (build 0x0019)

VERIFICATION

263 evidence files · 283 SHA-256 lines

57 benches · final_verify ALL CHECKS PASSED

Technical report 20 pages · full dossier 59 pages